

Benchmarking three-site water models for predicting phase equilibria in menthol-based hydrophobic eutectic solvents

Raj Akshat^a, Anand Bharti^{b,*}, Padmini Padmanabhan^{a,*}

^a Department of Bioengineering and Biotechnology, Birla Institute of Technology Mesra, Ranchi, Jharkhand 835215, India

^b Department of Chemical Engineering, Birla Institute of Technology Mesra, Ranchi, Jharkhand 835215, India

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ABSTRACT

Non-ionic hydrophobic eutectic solvents (ESs) are emerging as sustainable alternatives for water-intensive industrial applications, including separations and biomass processing. However, the predictive accuracy of molecular dynamics (MD) simulations for these systems depends critically on the choice of water model. In this study, we performed a systematic force field assessment of eight rigid three-site water models to evaluate their performance in describing a menthol: dodecanoic acid (3:1 M ratio) hydrophobic ES in contact with water. All models reproduced clear phase separation, but substantial variations were observed in water solubility, hydrogen bonding behaviour, interfacial tension (IFT), diffusion, and thermodynamic stability. The experimental solubility is ≈ 1.1 wt%, and among the tested models, OPC3 and TIP3P-ST showed the closest agreement. In contrast, TIP3P-FB significantly overestimated water uptake and predicted a stronger water-ES association. Interfacial tension varied widely across models, with higher effective dipole moment models such as OPC3 and SPC/E exhibiting larger IFT values and stronger exclusion of water from the ES phase. Umbrella sampling confirmed that transfer of a water molecule from the ES interior to the bulk phase is thermodynamically favourable for all models ($\Delta G < 0$). Models such as SPC/E and TIP3P-Ew exhibited higher free energy plateaus of approximately 75 to 80 kcal/mol, whereas TIP3P-FB showed a lower barrier of approximately 10 kcal/mol, enabling greater stabilisation of water within the ES phase. OPC3 and TIP3P-ST provide the most consistent description of phase equilibria and interfacial properties.

1. Introduction

The search for sustainable alternatives to conventional solvents is a central theme in green chemistry. However, most traditional solvents originate from finite petrochemical resources, tend to be highly volatile, and can present environmental and health risks because of their toxicity and limited biodegradability [1]. In this context, eutectic solvents (ESs) have emerged as promising candidates due to their low cost, ease of synthesis, tunable properties, and favourable environmental profiles. ESs are typically formed through the complexation of a hydrogen bond donor (HBD) and a hydrogen bond acceptor (HBA), resulting in a significant depression of the melting point compared to their components. This unique interaction allows ESs to remain liquid at or near room temperature, even when composed of solid precursors. Their structural similarity to ionic liquids (ILs), coupled with enhanced biodegradability and lower toxicity, makes ESs attractive for a range of applications, including extraction [2–4], catalysis [5,6], CO₂ capture [7,8], and

biomass processing [9,10].

Despite these advantages, ESs often suffer from high viscosity, which hinders mass transfer and limits their practical applicability. One widely adopted strategy to overcome this limitation is the addition of co-solvents, with water being the most common due to its low viscosity, polarity, abundance, and environmental compatibility. However, the introduction of water into ESs does more than provide macroscopic improvements, as it also alters key physicochemical properties of the mixture, which in turn collectively influence the solubilization and transport of target compounds. As Dai et al. (2015) showed, the addition of water to natural ESs significantly altered their viscosity, conductivity, polarity, and even their solubilization capacity for poorly soluble compounds like quercetin and carthamin [11]. The polarity shift induced by water addition can profoundly affect extraction performance. According to the principle of “like dissolves like,” matching the polarity of the ES-water mixture to that of the target analyte can enhance solubility [12]. However, as Ahmadi et al. [13] and other studies have noted, polarity

* Corresponding authors.

E-mail addresses: abharti@bitmesra.ac.in (A. Bharti), padmini@bitmesra.ac.in (P. Padmanabhan).

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alone does not dictate the extraction efficiency of ES-water mixtures. Hydrogen bonding, ionic interactions, and steric effects also play critical roles. Gabriele et al. reported that in choline chloride-ethylene glycol systems, the addition of water reduces viscosity while weakening the hydrogen bond interactions between the HBA and HBD [14]. At low concentrations, water integrates into the ES network and can enhance structural stability through additional hydrogen bonding [15]. Excessive hydration can disrupt the supramolecular architecture, as shown in studies on choline chloride-urea mixtures, where strong HBA-HBD interactions are progressively weakened, eventually leading to complete structural breakdown [16]. Choi et al. further observed that certain ESs retain water strongly and irreversibly, with bound water molecules that cannot be removed through evaporation, indicating substantial and irreversible alterations to the microstructure [17]. FTIR and NMR studies have confirmed that progressive water addition weakens HBA-HBD interactions and can lead to complete loss of structural integrity [11]. Additional investigations demonstrated that water content not only influences viscosity and polarity but also affects the rheological properties and moisture absorption capacity of ESs [18,19].

While hydrophilic ESs have been extensively studied, research attention has more recently shifted toward hydrophobic ESs, a class of solvents that was first explored in depth by van Osch et al. in 2015 [20]. These hydrophobic systems are typically composed of molecules with low polarity, such as long-chain fatty acids, menthol, thymol, ibuprofen, and certain organic acids [21–24]. Unlike their hydrophilic counterparts, hydrophobic ESs exhibit lower conductivity and higher viscosity. However, their immiscibility with water makes them especially attractive for extracting bioactive compounds and metal ions from aqueous environments [3,4,25–29]. Though still underexplored compared to traditional ESs, the physicochemical properties of hydrophobic ESs, including viscosity, density, vapor pressure, and structural behaviour, have been investigated using a combination of theoretical modeling and molecular dynamics simulations [4,30–33]. These studies provide valuable insights into their formation mechanisms, structural stability, and solvent capabilities, highlighting their potential as next-generation green solvents for challenging separation and catalytic processes.

In molecular dynamics studies, water models are used to simplify the representation of water molecules while still capturing their essential structural, dynamical, and thermodynamic properties. Over the years, a wide range of classical water models have been developed, differing in the number of interaction sites, the flexibility of their geometry, and the treatment of electrostatic interactions. Among these, rigid non-polarizable models such as TIP3P, SPC, TIP4P, and their variants are the most commonly used due to their balance between computational efficiency and physical accuracy. Three-site rigid models such as TIP3P [34–36] and SPC/E [37–40] are particularly popular for their efficiency and accuracy, and they have also been employed in studies involving water and hydrophobic ESs. Evidence from ionic liquid-water systems shows that the choice of water model can significantly influence simulation outcomes. For example, one study compared SPC/E and TIP5P models with both full-charge and scaled-charge IL force fields. The full-charge force field produced an interface for both models, but the predicted concentrations were higher than experimental values. Scaling the charges improved agreement with experiment [41]. In another related work, Kelkar et al. [42] tested various SPC-based water models with different partial charges to reflect varying electrostatic environments in the 1-ethyl-3-methylimidazolium ethylsulfate [Emim][EtSO₄]-water mixture. Using an IL force field with $\pm 1.0e$ charges, they observed that less polarised water models produced excess volume and enthalpy values closer to experimental data. These findings highlighted the importance of polarisation effects and suggested that polarizable models could further improve predictive accuracy.

By analogy, in hydrophobic ESs, the pronounced polarity mismatch between the ES and water can make simulation results highly sensitive to the choice of water model. However, systematic evaluations of water models in the context of hydrophobic ESs are scarce. In this study, we

address this gap by performing a comprehensive assessment of eight different three-site rigid water models for ES-water systems. Molecular dynamics simulations of binary mixtures of water and menthol-dodecanoic acid (3:1 M ratio) (MD (3:1)) were performed to evaluate how different water models influence the structural and dynamical properties. The force field assessment focuses on identifying the most suitable water model for accurately simulating MD (3:1), ensuring reliable predictions of phase separation, solvation behaviour, and hydrogen bonding dynamics in hydrophobic ES-water systems.

2. Computational details

The schematic representations of menthol, dodecanoic acid, and water are depicted in Fig. 1. The Optimized Potentials for Liquid Simulations-All Atom (OPLS-AA) force field was used to describe the interactions within the systems, capturing the energy contributions effectively. The force fields for menthol and dodecanoic acid were obtained from the LigParGen web server, which employs the 1.14*CM1A charge models to ensure accurate simulation parameterisation [43,44]. Eight different 3-point water models, namely, SPC, TIP3P, SPC/E, TIP3P-Ew, TIP3P-FB, SPC/ε, OPC3, and TIP3P-ST, were selected to study the binary liquid-liquid phase equilibria (Table 1) [45–52]. The bond lengths and angles were kept fixed in these water models. Fixed-point charges were assigned to each atom site, while only one Lennard-Jones interaction site was located on the oxygen atom. All simulations were conducted using GROMACS package version 2024.2 [53].

2.1. Force field validation

Firstly, molecular simulations were carried out for pure menthol: dodecanoic acid (3:1) eutectic solvent to validate the force field parameters. Three hundred (300) molecules of menthol and one hundred (100) molecules of dodecanoic acid were placed inside a cubic box of dimension $7 \times 7 \times 7 \text{ nm}^3$. The simulation consisted of energy minimisation using the steepest descent algorithm (10,000 steps) to eliminate atom overlapping. Subsequently, initial velocities were assigned from a Gaussian distribution at the specified temperature to ensure a realistic starting point. The simulation was initiated at a base temperature of 5 K and progressively raised to 400 K via a stepwise heating process over 500 ps in a canonical ensemble (NVT). The system temperature was then held at 400 K for 100 ps. In the cooling step, the system temperature was reduced to 300 K over 200 ps and held at 300 K for 100 ps. The heating was followed by the cooling, in which the system temperature was again raised to 400 K over 200 ps, then kept at 400 K for 100 ps. This cooling-heating cycle was repeated for 10,000 ps (10 ns). The temperature annealing process was adopted to facilitate proper mixing of the viscous systems. Thereafter, the system was equilibrated for 20 ns at 400 K and 1 atm pressure in an isothermal-isobaric (NPT) ensemble. The final configuration after a 20 ns simulation was taken as the starting point, and five different simulations in the NPT ensemble were carried out corresponding to the temperatures 298 K, 303 K, 308 K, 318 K, and 328 K for 20 ns.

The temperature during the NPT simulation was controlled by a Nose-Hoover thermostat [54], while the pressure was maintained using the Parrinello-Rahman barostat [55]. A time constant of 1 ps was used during the simulation for the applied thermostat and barostat. Periodic boundary conditions were set in all directions, with a 1.4 nm cutoff for calculating LJ and direct space electrostatic potential calculations. The particle mesh Ewald (PME) method was used to compute the long-range electrostatic interactions, where the 4th interpolation order was considered for higher accuracy. All the bonds involving H-atoms were kept rigid using the LINCS algorithm [56]. Newton's equation of motion was integrated using the leap-frog algorithm with a time step of 1.0 fs. All the analyses from the NPT ensemble were carried out during the last five (05) ns.

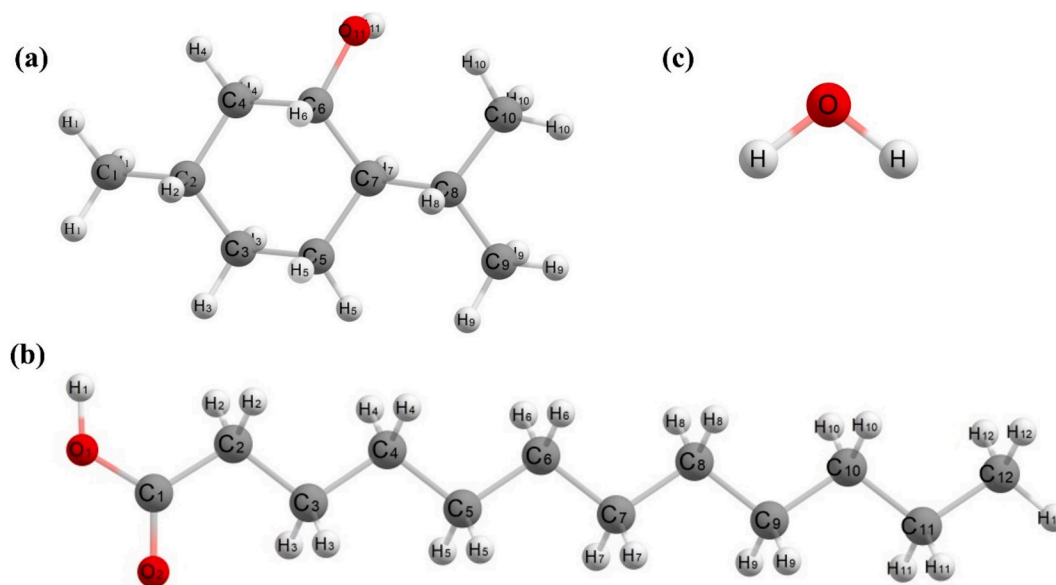


Fig. 1. Schematic representation of molecules studied: (a) Menthol (MEN), (b) Dodecanoic acid (DOD), and (c) Water (HOH).

Table 1

Force Field Parameters of 3-Point Water Models.

Water model	H-charge (e)	O-charge (e)	O (σ) (Å)	O (ϵ) Kcal/Mol	O-H (Å)	H-O-H (°)	Year published
TIP3P-ST	0.425560	-0.85112	3.19257	0.143858	1.02300	108.1100	2019
OPC3	0.447585	-0.89517	3.17427	0.163406	0.97888	109.4700	2016
SPC/c	0.445000	-0.89000	3.17850	0.168704	1.00000	109.4500	2015
TIP3P-FB	0.424220	-0.84844	3.17800	0.155425	1.01180	108.1500	2014
TIP3P-Ew	0.415000	-0.83000	3.18800	0.102000	0.95720	104.5200	2004
SPC/E	0.423800	-0.84760	3.16600	0.155350	1.00000	109.4667	1987
TIP3P	0.417000	-0.83400	3.15061	0.152100	0.95720	104.5200	1983
SPC	0.410000	-0.82000	3.16600	0.155350	1.00000	109.4667	1981

2.2. Molecular dynamic simulation of binary mixture (water/ES)

A binary mixture of eutectic solvent and water was studied following a pure component simulation study. A three-box strategy was employed to pack the molecules for binary system simulations. The simulation box obtained from the pure eutectic solvent study was extended along the X-axis (both directions) to a cuboidal shape measuring $15 \times 5 \times 5 \text{ nm}^3$. The eutectic solvent (300 molecules of menthol and 100 molecules of dodecanoic acid) was kept in the middle, while 3000 water molecules were placed on both sides. Periodic boundary conditions (PBC) were applied along all three dimensions to mimic an infinite system. Interatomic forces were modeled using the Lennard-Jones 6–12 potential for van der Waals interactions, and long-range electrostatics were calculated using the particle mesh Ewald (PME) method [57]. Both interactions employed a cutoff radius of 1.2 nm. Energy minimisation of the system was performed using the steepest descent algorithm. The system dynamics were then simulated using the velocity-verlet integration algorithm, with a timestep of 1.0 fs [58,59]. After minimisation, a 10 ns simulation was conducted in the NVT ensemble at 303 K, followed by a 20 ns NpT simulation at 303 K. For the production phase, an additional 50 ns simulation was run in the NpT ensemble at 303 K. During these simulations, temperature control was achieved using the Nose-Hoover thermostat with a 5 ps time constant [54,60], while pressure was maintained with the Parrinello-Rahman barostat, also using a 5 ps time constant [55].

2.3. Umbrella sampling and free energy calculation

Umbrella sampling calculations were performed to determine the free energy change (ΔG_{wat}) for transferring a water molecule from the eutectic solvent (ES) phase to the bulk water region. Initial configurations were taken from the equilibrated NpT trajectory of the binary mixture. A water molecule that had naturally diffused into and remained within the ES domain during equilibration was selected to ensure that the process represents removal under equilibrium conditions. The reaction coordinate was defined as the distance between the centre of mass (COM) of the ES and the COM of a selected water molecule initially located within the ES domain. The pathway was divided into 25 evenly spaced windows extending from the ES interior to the bulk region. In each window, the water molecule was restrained near the target position using a harmonic umbrella potential applied through the GROMACS pull code. A force constant of $100 \text{ kcal/mol } \text{\AA}^{-2}$ was used to ensure adequate sampling and overlap between adjacent windows.

Each window was equilibrated and subsequently simulated for 1.0 ns in the NpT ensemble, followed by 2 ns of NpT ensemble at 303.15 K. The restrained coordinate was recorded throughout the simulations. The biased distributions from all windows were unbiased and combined using the Weighted Histogram Analysis Method (WHAM) implemented in gmx wham [61,62]. The potential of mean force was constructed using 100 bins with a convergence tolerance of 10^{-5} , yielding the free-energy profile for water transfer from the ES phase to the bulk environment.

Further, to determine the self-diffusion coefficient (D), an independent simulation of 50 ns was utilized. Radial distribution functions

(RDF) were analyzed to understand molecular interactions within the system. Hydrogen bonding was examined using the Visual Molecular Dynamics (VMD) tool, providing insights into interactions between the ES and the mentioned water models [63]. Local mass density profiles and interfacial tension calculations also offered further information on system behaviour and phase characteristics.

3. Results and discussion

3.1. Density estimation

Before conducting the binary LLE study using molecular simulations, the density of MD (3:1) was calculated in the isothermal-isobaric ensemble. The simulation-based density estimation (with ± 0.8 charge scaling factor) and its comparison to experimental data served as the primary validation for the force field employed in this work. This scaling approach has been widely adopted in simulations of ionic liquids [64,65] and deep eutectic solvents [66,67]. In our previous work, this scaling factor was systematically validated across chemically diverse ES compositions at 298.15 K, where the ± 0.8 scaled force field consistently yielded lower deviations from experimental densities compared to the full ± 1.0 charge parameterisation [4]. For the MD 3:1 system specifically, the %ARD reduces from 1.679% with full charges to just 0.336% with ± 0.8 scaling. Also, simulated densities of MD (3:1) across various temperatures in Table 2 showed excellent agreement with the experimental results. Minor deviations were observed, with the maximum and minimum absolute relative deviations being 0.58% and 0.29%, respectively.

3.2. Local mass density profile and interfacial tension

The local mass density profile of the ES and various water models offers key insights into the phase behaviour and interaction dynamics at the ES/water interface. As shown in Fig. 2, the interface was identified as the region where the density gradually changed from the ES phase to the water phase. For consistency, its thickness was estimated as the X-distance between the points where each phase's density reached 90% of its bulk value [68]. The analysis revealed that the ES remained largely intact, without entering the water phase. The interface thickness is consistent across most water models, ranging from 1.16 to 1.24 nm on the right interface and 1.14–1.23 nm on the left interface, as shown in the computed data presented in Table 3. Fig. 3 further complements this analysis by presenting the density profiles of the individual components of ES (menthol and dodecanoic acid) with different water models. The blue shaded area highlights the interface. Since dodecanoic acid is more hydrophilic than menthol, as indicated by the density profiles, having a higher proportion of menthol compared to dodecanoic acid (with a molar ratio of 3:1) could result in increased interfacial tension and greater overall hydrophobicity compared to systems with a lower menthol content. This analysis emphasizes the hydrophobic nature of ES, as neither menthol nor dodecanoic acid leaches into the water phase. This observation aligns well with previous results, confirming the

stability and separation of the ES phase from water. By analyzing the local mass density profiles, this study provides a comprehensive understanding of how various water models behave at the ES/water interface. The analysis confirms that ES remains stable and intact, with no leaching of its components into the water phase, demonstrating the consistent hydrophobic behaviour of ES across all water models.

Further, the interfacial tension (γ) plays a crucial role in the stability and dynamics of the liquid-liquid interface, significantly influencing phase separation and the dispersion of one phase into another. The interfacial tension (IFT), γ , was determined from the pressure tensor components (P_{xx} , P_{yy} , and P_{zz}) following eq. 1.

$$\gamma = \frac{1}{2} \left[L_x \left(P_{xx} - \frac{P_{yy} + P_{zz}}{2} \right) \right]_t \quad (1)$$

In this equation, L_x denotes the length of the simulation box in the X-direction, which is perpendicular to the Z-Y plane. The factor of $\frac{1}{2}$ accounts for the two interfaces present in the simulation box, and the notation $[\dots]_t$ represents an ensemble average over the simulation time [69]. The final IFT values were computed as ensemble averages over a 100 ns simulation. The associated errors in γ were evaluated using the standard error derived from dividing the 100 ns simulation into ten 10 ns blocks. The calculated interfacial tensions for various water models are presented in Table 4. The OPC3 model exhibited the highest interfacial tension (30.26 mN/m), indicating a more rigid interface, which would limit water dispersion into the ES phase. This high value suggests that OPC3 may better maintain a clear separation between the phases, resulting in lower water solubility in the ES. Similarly, models like TIP3P-ST, SPC/ε, and TIP3P-FB displayed relatively high interfacial tension values of 22.04, 24.76, and 25.66 mN/m, respectively. These values suggest that these models exhibit strong interfacial interactions, albeit not as pronounced as OPC3. Such strong interfaces indicate less intermixing and higher phase stability. On the lower end, TIP3P and SPC show the lowest interfacial tension values, 17.38 mN/m and 17.86 mN/m, respectively. These values suggest comparatively higher water-ES interactions, allowing more water molecules to penetrate the ES phase. Lower interfacial tension suggests a more fluid and less resistant interface, promoting greater interaction between the two phases and potentially resulting in greater mixing or solubility. An interesting observation is that the OPC3 model assigns the highest negative charge to the oxygen atom (−0.89517e) among all 3-point models, whereas TIP3P (−0.83400e) and SPC (−0.82000e) have the lowest oxygen charges. This difference in charge magnitude influences the electrostatic interactions of water molecules, both with each other and with other molecules. A more negative oxygen charge enhances the water dipole moment, strengthens the magnitude of hydrogen bonding between water molecules, and thus may lead to higher interfacial tension compared to TIP3P and SPC. A simulation using TIP3P and SPC would suggest a relatively fluid, low-cohesion interface that could be prone to emulsification and slow phase separation, potentially leading to oversized settler designs. This clearly highlights how the choice of force field can influence water's interfacial behaviour. Although all models preserve the same qualitative phase separation, the interfacial tension values differ by nearly 70% across models, directly demonstrating that water model choice produces clear and measurable quantitative differences in interfacial cohesion.

3.3. Water solubility in the extract phase

In this study, water uptake into the extract phase was analyzed across various water models, considering the influence of the interface between the phases. The analysis was conducted using three different approaches: (a) considering water molecules in the bulk region and excluding those at the interface (W_{WOF}), (b) including water molecules located at the midpoint of the interface (W_{MIF}), and (c) accounting for water molecules present across the entire interfacial region (W_{WIF}).

Table 2

Comparison between simulated (with ± 0.8 charge scaling factor) and experimental density of MD (3:1).

Temp (K)	Simulated density (kg/m ³)	Experimental density (kg/m ³)	%ARD
298	0.896 ± 0.002	0.8930	0.34
303	0.892 ± 0.003	0.8894	0.29
308	0.891 ± 0.002	0.8859	0.58
318	0.883 ± 0.005	0.8787	0.49
328	0.874 ± 0.002	0.8714	0.30
		%AARD	0.40

$$\%ARD = \frac{|Experimental\ density - simulated\ density|}{Experimental\ density} \times 100.$$

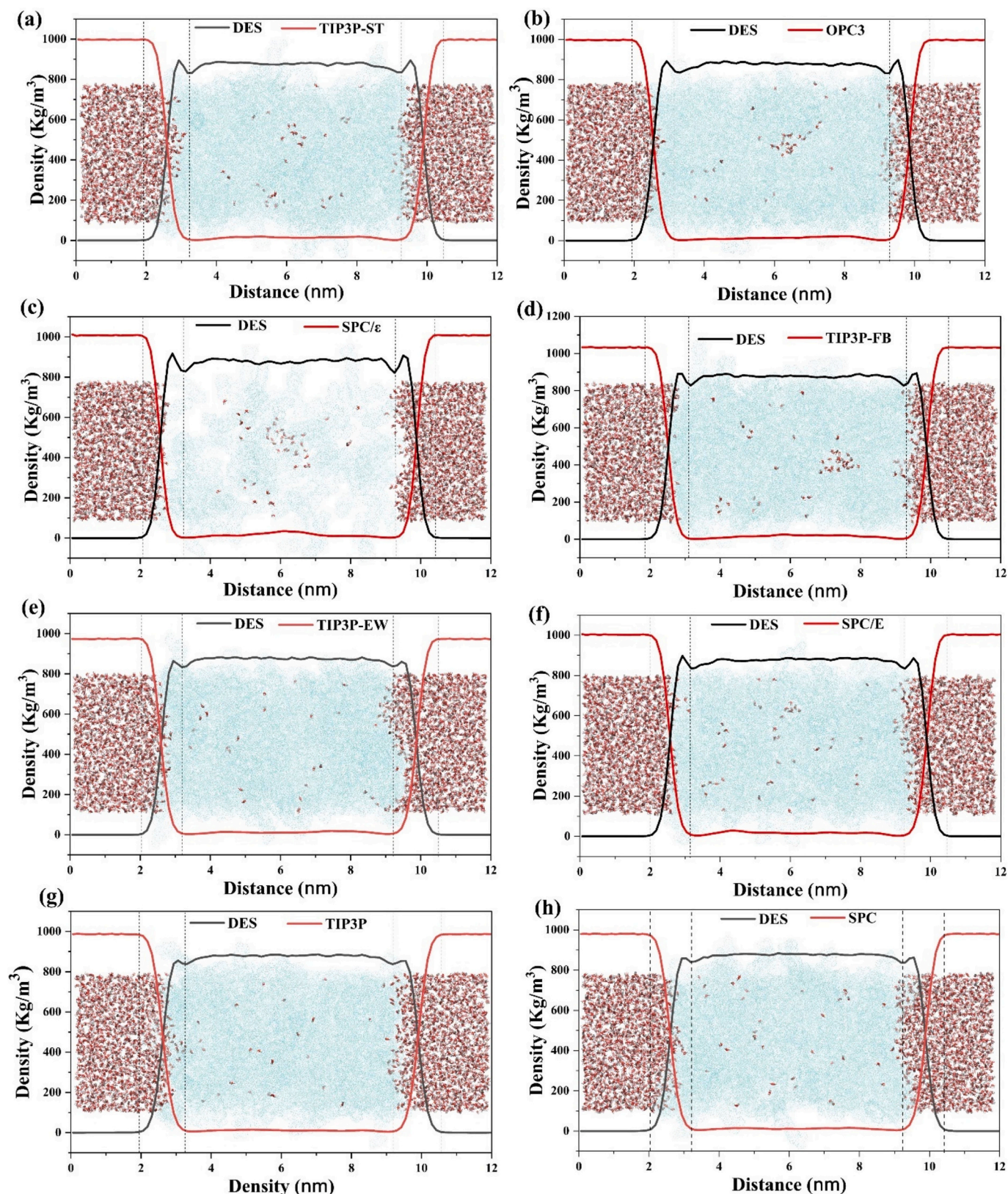


Fig. 2. Local mass density profiles and interface of the ES/water system with various water models at 303 K and 1 bar.

Table 5 presents the weight percentage (wt%) of water in the extract phase for each water model.

The number of water molecules entering the extract phase (bulk) without accounting for the interface (N_{WOIF}) varies across models,

ranging from 19 in the TIP3P model to 71 in TIP3P-FB, with the corresponding weight percentages being generally low, the highest being 7.10% for TIP3P-FB. When considering the midpoint of the interface (N_{MIF}), more water molecules penetrate the extract phase, with TIP3P-

Table 3

Interfacial width of the ES/water phase measured at the right and left interfaces using different water models.

Water model	Interfacial width (nm)	
	Right	Left
TIP3P-ST	1.1800	1.1400
OPC3	1.2400	1.1500
SPC/c	1.2200	1.1600
TIP3P-FB	1.2400	1.1600
TIP3P-Ew	1.2200	1.1900
SPC/E	1.1600	1.1800
TIP3P	1.2000	1.2300
SPC	1.1600	1.2200

FB, SPC/E, and TIP3P-ST showing a significant increase, yielding weight percentages of 19%, 11.01%, and 10.41%, respectively. For the complete interface case (N_{WIF}), all models display a notable rise in water content within the extract phase. TIP3P-FB and SPC/E models exceed 20 wt% water content, with TIP3P-FB reaching the highest value of 26.79%. This nearly fourfold difference directly demonstrates that water models do not behave equivalently and that they predict markedly different levels of water penetration into the ES phase.

The Karl Fischer titration result showed a water content of 1.1% in the extract phase, as reported for the same system in the literature [4]. Among the simulated models, TIP3P (wt% = 2.00), OPC3 (wt% = 2.11), and TIP3P-Ew (wt% = 2.42) closely match the experimental values, particularly when considering the water molecules in the bulk region and excluding those at the interface. The TIP3P-FB model, while capturing the highest water content in all cases, appears to overestimate the experimental results, especially in the case when water molecule accounting was done considering water presence across the entire interfacial region. The progressive rise in water content from N_{WOIF} to N_{WIF} indicates that the interface plays a crucial role in water solubility within the solvent, where water molecules are more likely to enter and interact with the ES at the interfacial region. A simulation framework built on TIP3P-FB would therefore systematically overestimate the degree of water contamination in the extract phase, leading to overly conservative process designs with unnecessarily high solvent-to-feed ratios or additional downstream drying steps that may not be required in practice. Models like TIP3P, OPC3, and TIP3P-Ew, which approach the experimental value of 1.1%, better capture this balance of water migration across the interface, aligning more closely with experimental observations. Thus, the solubility results are influenced not only by the choice of water model but also by the method used to count the number of water molecules present in the solvent phase.

3.4. Structural analysis

Molecular dynamics simulations were used to calculate the RDF to investigate the interactions between water and the eutectic solvent. The eq. 2 describes the RDF.

$$g(r) = \frac{\langle \rho(r) \rangle}{\rho} \quad (2)$$

Where $\rho(r)$ represents the local particle density at a distance r , and ρ is the average particle density of the system. This function helps in understanding the probability of finding a particle at a certain distance from a reference particle, thereby providing insights into the structural arrangement within the system.

The coordination number (CN) was then calculated by integrating the RDF ($g(r)$) up to the first solvation shell, which corresponds to the first local minimum of the RDF curve as shown in eq. 3.

$$CN = 4\pi\rho \int_0^{R_{min}} r^2 g(r) dr \quad (3)$$

Here, $4\pi^2 dr$ represents the volume of a spherical shell, and the integration is performed from 0 to R_{min} , the position of the first minimum. This method allows for a quantitative evaluation of particle arrangement, highlighting how different water models affect the coordination number with MD(3:1).

Several atom-atom RDFs using the OPC3 model were computed to map the possible interactions between the ES components and water molecules, as shown in Fig. 4. The RDFs in Fig. 4 exhibit a sharp and well-defined first peak at approximately 0.20–0.22 nm for the H_{OH} (dodecanoic acid) - O_{water} pair. This distance corresponds to the typical O-H...O hydrogen-bond separation, confirming that water forms a structured first solvation shell around the acidic -OH group of dodecanoic acid. In contrast, menthol-water RDFs display no distinct first peak, and $g(r)$ remains close to zero for $r < 0.3$ nm, indicating the absence of a well-defined solvation shell and confirming that menthol does not engage in hydrogen bonding with water.

From these observations, we concluded that menthol plays a significant role in imparting hydrophobicity to the ES, while dodecanoic acid is primarily responsible for interacting with water through hydrogen bonding. Accordingly, further investigation was carried out on the interactions between the hydrogen atom of the -OH group in dodecanoic acid and the oxygen atom of water, using RDF and CN analyses across different water models.

Fig. 5 presents the analysis of RDF and CN for different water models interacting with ES, revealing a CN range between 0.44 and 0.61. TIP3P-FB had the highest CN (0.605), indicating stronger interactions with ES and suggesting significant hydrogen bonding. In contrast, the TIP3P model had the lowest CN (0.441), showing comparatively weaker interaction. Other models like OPC3 (0.522) and SPC/E (0.552) displayed intermediate interactions.

3.5. Hydrogen bond analysis

The analysis of hydrogen bond interactions using various water models reveals key insights into their behaviour in a hydrophobic eutectic solvent phase. In such a hydrophobic environment, the potential for hydrogen bonding between water and the ES is naturally constrained due to the nonpolar nature of the solvent. However, despite these limitations, the hydrogen bonding data reveal notable variations in the strength of interactions across different water models. To quantify these interactions, the cutoff criteria for hydrogen bond length and angle were determined based on the geometrical criterion, with the H-bond length set at 3.5 Å and the angle at 30° [70,71]. Fig. 6 (a) illustrates the average number of hydrogen bonds (H-bonds) formed between the ES and various water models. Water models such as TIP3P-FB (2.19) demonstrate the highest average H-Bond counts, indicating that these models facilitate stronger water-water or localized water-ES interactions. This suggests that these models promote better integration of water molecules into the ES phase, even in a predominantly hydrophobic setting.

In contrast, models such as TIP3P (1.73) and TIP3P-EW (1.83) exhibit significantly lower average hydrogen bond counts. These models' reduced hydrogen bonding potential corresponds to weaker interactions with the ES phase, resulting in less water incorporation and less stable molecular structures within the hydrophobic environment. Additionally, as observed in prior studies, water solubility in ES systems increases with the number of hydrogen bonds formed. This reinforces the role of hydrogen bonding strength as a critical factor in determining water solubility in hydrophobic ES environments.

To quantitatively assess the hydrogen bond dynamics of various water models in the ES phase, we computed the hydrogen bond autocorrelation function ($c(t)$) as shown in eq. 4. The $c(t)$ provides a measure of the time-dependent behaviour of hydrogen bonding, capturing the breaking and reforming of hydrogen bonds over time [72]. This analysis follows the theoretical framework established by Luzar and Chandler, allowing for a detailed examination of hydrogen bond lifetimes.

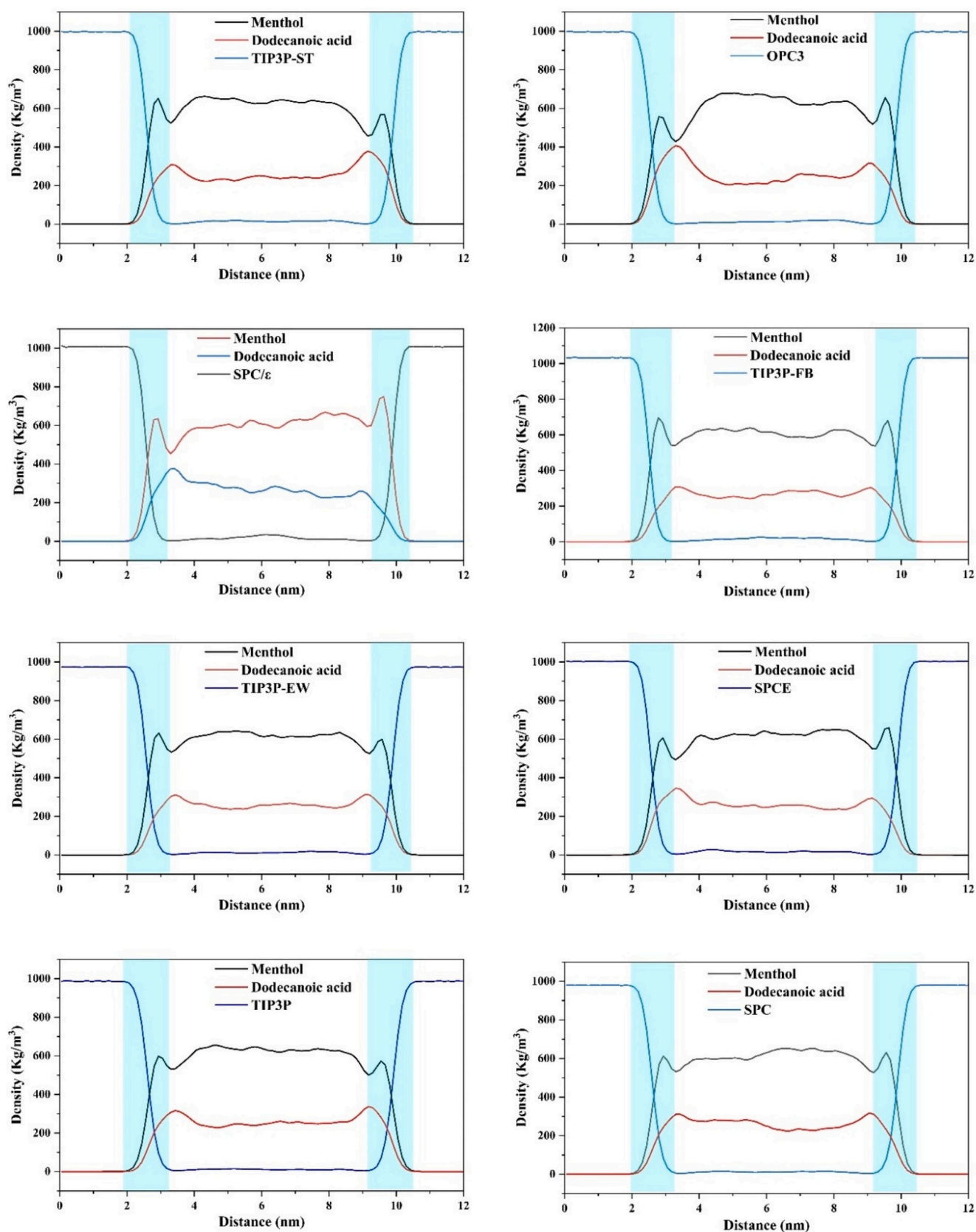


Fig. 3. Local mass density profiles of menthol, dodecanoic acid, and water for all eight water models at 303 K and 1 bar. The shaded region marks the interfacial zone, where the density transitions between the ES phase and the water phase, indicating the interfacial width.

Table 4

Computed interfacial tension of the ES/water binary system (Menthol and dodecanoic acid in a 3:1 M ratio) with different water models at 303 K and 1 atm.

Water models	γ (mN/m)
TIP3P-ST	22.04 ± 0.86
OPC3	30.26 ± 0.91
SPC/ε	24.76 ± 0.86
TIP3P-FB	25.66 ± 0.87
TIP3P-Ew	20.74 ± 0.79
SPC/E	19.66 ± 0.72
TIP3P	17.38 ± 0.68
SPC	17.86 ± 0.66

Table 5

Distribution of water molecules in the extract phase using different water models. N_{WOIF} , N_{MIF} , and N_{WIF} represent the number of water molecules in the extract phase (a) excluding the interface (bulk region only), (b) at the midpoint of the interface, and (c) across the entire interfacial region, respectively. W_{WOIF} , W_{MIF} , and W_{WIF} denote the corresponding weight percentages of water in the extract phase for each counting approach.

Water models	Number of water molecules in the extract phase			Wt (%)		
	Without Interface (N_{WOIF})	Mid Interface (N_{MIF})	With Interface (N_{WIF})	W_{WOIF}	W_{MIF}	W_{WIF}
TIP3P-ST	33	108	183	3.43	10.41	16.45
OPC3	20	87	191	2.11	8.56	17.05
SPC/ε	41	71	164	4.23	7.10	15.00
TIP3P-FB	71	218	340	7.10	19.00	26.79
TIP3P-Ew	23	82	177	2.42	8.11	16.00
SPC/E	43	115	234	4.42	11.01	20.12
TIP3P	19	64	153	2.00	6.44	14.14
SPC	31	72	176	3.23	7.19	15.92

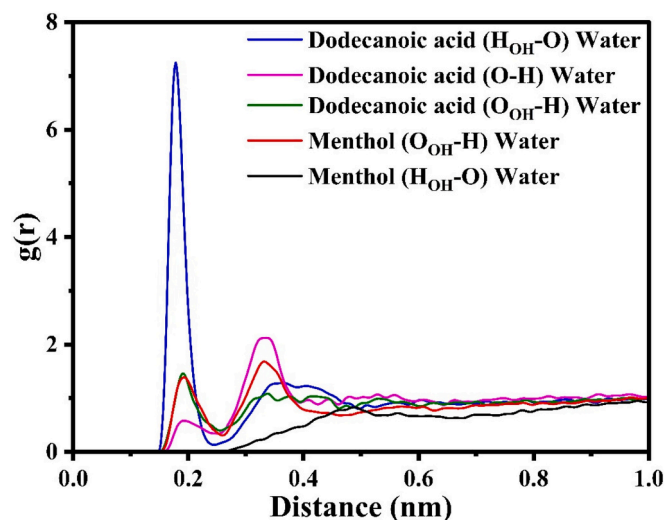


Fig. 4. Computed RDF plot between water and different functional groups of ES.

$$c(t) = \frac{\langle h(0)h(t) \rangle}{\langle h \rangle} \quad (4)$$

Where $h(t)$ is a descriptor that equals 1 when a hydrogen bond exists between a specific pair of molecules at time t , and 0 otherwise. The numerator, $\langle h(0)h(t) \rangle$, measures the likelihood that a hydrogen bond

present at $t = 0$ still exists at time t . The denominator, $\langle h \rangle$, normalizes the function by the average hydrogen bond population. The hydrogen-bond correlation function, depicted in Fig. 6 (b), illustrates the decay of hydrogen bond correlations over time for the different water models. By applying a tri-exponential decay model, the autocorrelation function was fitted according to eq. 5, with normalisation constraint $a_3 = 1 - a_1 - a_2$.

$$\tau_{\text{HB}} = a_1 \exp\left(-t/\tau_1\right) + a_2 \exp\left(-t/\tau_2\right) + a_3 \exp\left(-t/\tau_3\right) \quad (5)$$

The three characteristic time constants capture distinct water-exchange behaviours. The short- and intermediate-time components (τ_1 and τ_2) corresponds to rapidly exchanging and weakly associated water molecules, whereas the long-time component (τ_3) reflects the residence of strongly bound water molecules that remain within the first hydration shell of the eutectic solvent, indicative of slower exchange dynamics. The average hydrogen-bond lifetimes τ_{HB} are summarised in Table 6. Simulations by Liu et al. (2018) have shown that the average hydrogen bond lifetime in liquid water occurs on two distinct timescales: a fast timescale of 0.78 ps and a slower one of 5.11 ps [73]. Our findings reveal that the overall hydrogen bond dynamics in ES solutions are significantly slower compared to pure water systems. This can be attributed to the unique interactions between water and ES components, which hinder the rapid breaking and reforming of hydrogen bonds typically observed in bulk water.

The data in Table 6 further highlight variations in hydrogen bond lifetimes across different water models. Notably, the TIP3P-FB model exhibits the longest hydrogen bond lifetime at 15.89 ps, indicating the most stable hydrogen bond network in the ES environment. In contrast, the TIP3P model shows the shortest lifetime at 9.32 ps, suggesting more transient hydrogen bonds. Other models, such as SPC/E and OPC3, display intermediate hydrogen bond lifetimes, reflecting the balance between stability and flexibility in their hydrogen bonding behaviour. The variability in hydrogen bond lifetimes across these water models emphasizes the importance of water model selection when studying ES systems' solvation and dynamic properties.

3.6. Interaction energy parameters

Interaction energy was studied to assess how different water models represent the association between the hydrophobic ES and water. The total non-bonded interaction energy was calculated as the sum of van der Waals and electrostatic contributions, as expressed in eq. (6).

$$E_{\text{non-bonded}} = E_{\text{van der waals}} + E_{\text{electrostatic}} \quad (6)$$

The total interaction energies between the hydrophobic ES and the tested water models range from approximately $-1.80 \text{ kJ mol}^{-1}$ (TIP3P-Ew) to $-2.48 \text{ kJ mol}^{-1}$ (TIP3P-FB) as observed from Fig. 7. In all cases, the Coulombic contribution dominates over the Lennard-Jones term, indicating that the primary interactions at the ES-water interface are electrostatic in nature, despite the overall hydrophobic character of the ES. The electrostatic ordering at the interface is the primary factor restricting water mobility.

A clear relationship is observed between the magnitude of interaction energy and the extent of water entry into the ES phase. TIP3P-FB shows the highest interaction energy magnitude ($-2.48 \text{ kJ mol}^{-1}$) with a Coulombic term of about $-2.15 \text{ kJ mol}^{-1}$, suggesting greater penetration of water into the ES matrix. In contrast, TIP3P-Ew displays the weakest total interaction ($-1.80 \text{ kJ mol}^{-1}$) and is therefore considered to underpredict the ES-water interactions for this system. This underprediction reflects limited water penetration and reduced interfacial structuring. TIP3P ($-2.12 \text{ kJ mol}^{-1}$) and SPC ($-2.15 \text{ kJ mol}^{-1}$) exhibit intermediate values and can be considered reasonable in their predictions. The consistency between solubility and interaction energy trends reinforces the conclusion that greater water entry into the ES is directly associated with stronger interactions in molecular

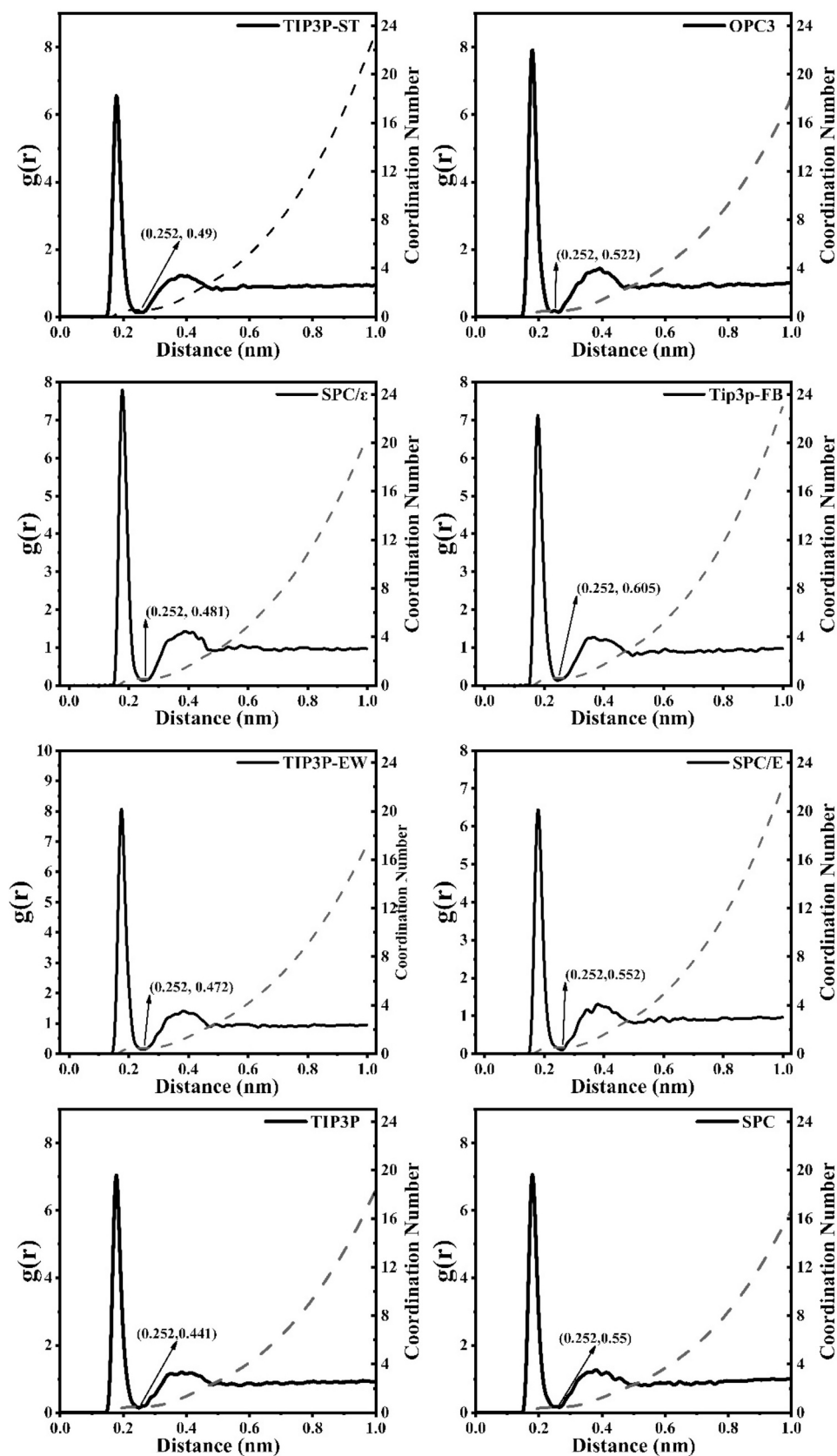


Fig. 5. RDF and CN of the 'O' atom of different water models with the hydroxyl hydrogen (H_{OH}) in Dodecanoic Acid of ES (MD 3:1).

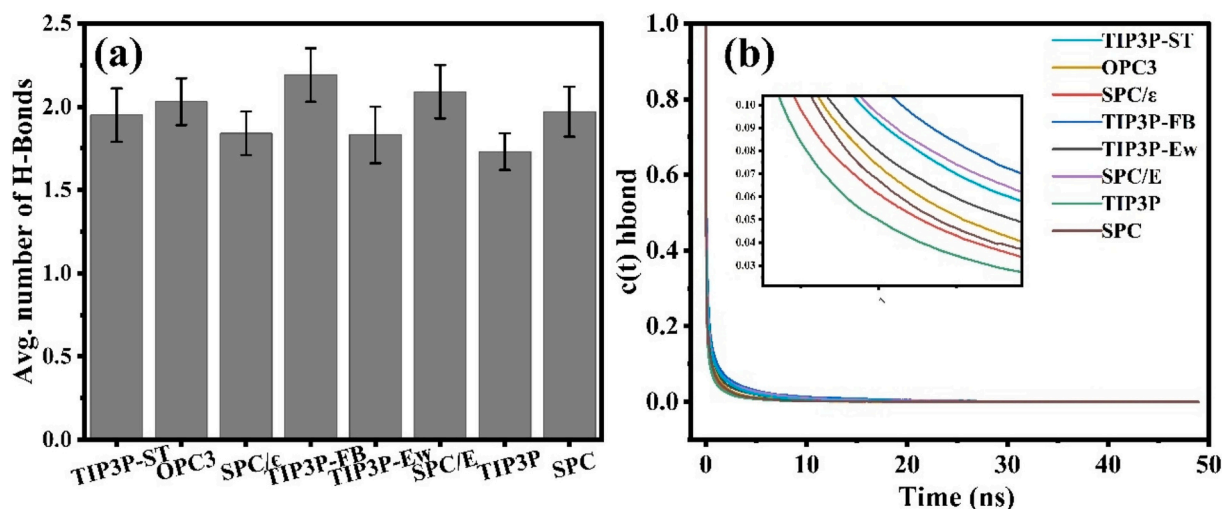


Fig. 6. (a) The average number of hydrogen bonds (per ES molecule) formed between ES and different water models at 303 K. (b) Hydrogen-bond correlation function $c(t)$ for the water molecules in the ES phase.

Table 6

Average hydrogen bond lifetime between water and ES using different water models.

Water models	τ_{HB} (ps)
TIP3P-ST	12.98 ± 0.67
OPC3	13.88 ± 0.83
SPC/ε	9.87 ± 0.92
TIP3P-FB	15.89 ± 1.11
TIP3P-Ew	11.88 ± 0.83
SPC/E	13.97 ± 0.57
TIP3P	9.32 ± 0.79
SPC	10.33 ± 0.43

dynamics simulations.

3.7. Self-diffusion coefficient

The self-diffusion coefficient was studied to understand the dynamics of the system's components. We calculated the mean-square displacement (MSD) of water using the eq. 7.

$$MSD = \frac{1}{N} \left[\sum_{i=1}^N |r_i(t) - r_i(0)|^2 \right] \quad (7)$$

In this equation, $r(t)$ and $r(0)$ represent the positions of atom i at times t and 0, respectively. The angle brackets $[\]$, indicates an average over all particles. The self-diffusion coefficient D , was determined from the MSD using the Einstein relation (eq. 8).

$$D_s = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \left(\sum_{i=1}^N |r_i(t) - r_i(0)|^2 \right) \quad (8)$$

Where N is the total number of particles, and the $1/6$ factor accounts for the three-dimensional nature of the system. This analysis allowed us to assess the diffusion and dynamics of water within the system.

Fig. 8 shows the MSD profiles for the different water models in the ES phase. The slope of each MSD curve is directly proportional to the diffusion coefficient, where a steeper slope signifies higher molecular mobility and weaker interactions with the surrounding solvent. In contrast, a gentler slope corresponds to slower diffusion, typically associated with stronger interactions or higher solubility affinity within the solvent matrix [25]. From the MSD trends, TIP3P and TIP3P-Ew display the steepest slopes, indicating greater water mobility, while

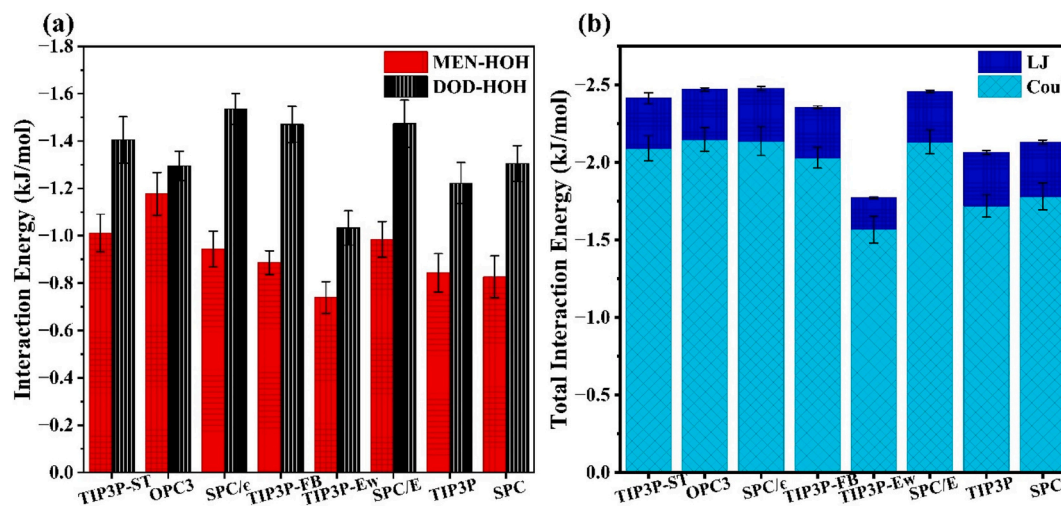


Fig. 7. (a) Interaction energy between MEN and DOD of ES, and, (b) Coulombic and Lennard-Jones interaction energies (in kJ mol^{-1}) between ES and different water models.

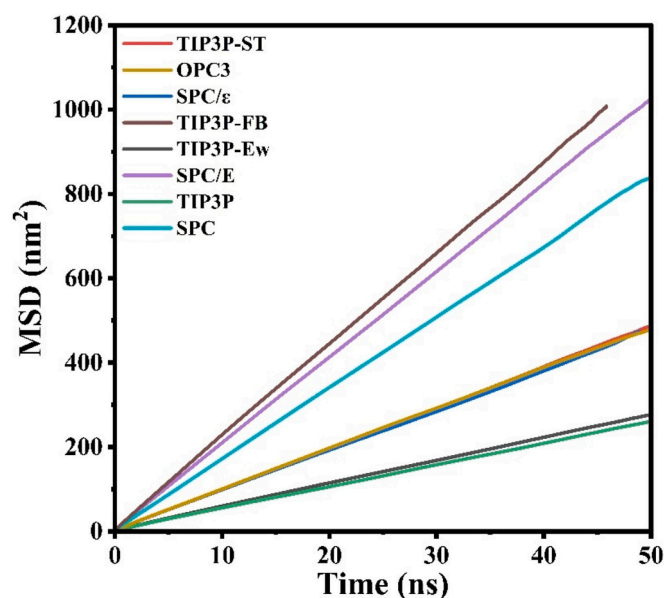


Fig. 8. MSD plot of different water models in the ES phase.

TIP3P-FB and SPC/E show the shallowest slopes, consistent with restricted molecular motion due to stronger water-ES interactions. Using Eq. 7, the diffusion coefficients were obtained from the MSD slopes, revealing the relative mobility of water molecules in the extract phase. The values in Table 7 reinforce the same performance trends observed earlier. The results demonstrate a range of diffusion coefficients for the various water models. TIP3P and TIP3P-Ew exhibited the highest diffusivity values, indicating greater mobility of water molecules in these models. On the other hand, the TIP3P-FB and SPC/E water models had the lowest diffusion coefficient, suggesting more restricted movement of water molecules in the extract phase. These findings reveal that the choice of water model can significantly influence the diffusivity in solvent environments, providing important insights into the transport properties and dynamics of water in such systems. In the earlier analysis, it was noted that the solubility of water increases with the strength of hydrogen bonding in the ES phase. This observation aligns with the current findings on the self-diffusion coefficients. Water models like SPC/E, which exhibited the lowest diffusion coefficient, likely form stronger hydrogen bonds within the ES, leading to more constrained movement. Conversely, models like TIP3P and TIP3P-Ew, which showed the highest diffusion coefficients, may experience weaker hydrogen bonding interactions in the ES, allowing for faster water movement. Thus, the interplay between hydrogen bonding and self-diffusion shows that models with stronger water-ES interactions tend to have lower diffusivity, while models with weaker interactions exhibit higher mobility.

Table 7
Self-diffusion coefficient (D) of water in the extract phase at 303 K and 0.1 MPa.

Water models	D (cm ² /s) × 10 ⁻⁵
TIP3P-ST	2.6173 ± 0.0416
OPC3	2.7958 ± 0.0139
SPC/ε	2.5974 ± 0.0244
TIP3P-FB	0.9055 ± 0.0061
TIP3P-Ew	3.4092 ± 0.0800
SPC/E	0.8520 ± 0.0142
TIP3P	3.5907 ± 0.0558
SPC	1.5595 ± 0.0364

3.8. Potential of mean force analysis

The hydrophobic character of the ES exhibits strong sensitivity to the chosen water model, as reflected in both the PMF profiles and the calculated transfer free energies ($\Delta G_{ES \rightarrow \text{water}}$) as shown in Fig. 9. Because all ΔG values are negative, transfer from the ES interior to bulk water is thermodynamically favourable across all models, confirming the hydrophobic nature of the ES phase. Ranking based on the magnitude of ΔG reveals that SPC/E and TIP3P-Ew predict the strongest water exclusion, exhibiting the most negative transfer free energies of ≈ -37 kcal/mol. These are followed by TIP3P-ST and OPC3, which also display substantial thermodynamic driving forces for expulsion. Standard TIP3P predicts a comparatively weaker preference for migration into bulk water, whereas TIP3P-FB exhibits the smallest magnitude of ΔG , ≈ -14 kcal/mol, indicating the greatest relative stabilisation of water within the ES phase among the models examined.

The divergence among the models arises from differences in their force field parameterisation, particularly in the balance between electrostatic and dispersion interactions. Models such as SPC/E and TIP3P-Ew incorporate enhanced effective dipole moments through their charge distributions, utilising oxygen partial charges of $-0.8476e$ and $-0.830e$, respectively. These parameters strengthen intermolecular hydrogen bonding and cohesive energy in bulk water. When transferred into the low dielectric, hydrogen bond-disrupted interior of the hydrophobic ES, these models incur a larger energetic penalty due to the loss of favourable water-water interactions, resulting in elevated PMF plateaus of approximately 75 to 80 kcal/mol. In contrast, reparameterized variants such as TIP3P-FB employ modified charge distributions, with $q(O) = -0.84844e$, and Lennard-Jones parameters, $\sigma = 3.178 \text{ \AA}$ and $\epsilon = 0.155865 \text{ kcal/mol}$, that rebalance electrostatic and dispersion contributions. This reduces effective overbinding in bulk water and lowers the free energy cost associated with cavity formation and confinement within the ES phase, leading to smaller PMF barriers and more moderate ΔG values.

The PMF profiles further illuminate the structural transition between phases. Although peak PMF heights within the ES interior vary significantly, ranging from ≈ 10 kcal/mol for TIP3P-FB to approximately 80 kcal/mol for TIP3P-Ew, these barrier magnitudes do not strictly correlate with the overall transfer free energies. This distinction underscores the difference between interfacial crossing barriers and bulk thermodynamic stability. At the interface, $z \approx 0.5$ to 1.0 nm , models such as TIP3P-ST, TIP3P-FB, and SPC exhibit a shallow local minimum, indicative of a metastable adsorption state in which water molecules are transiently stabilized at the ES surface before diffusing into bulk water. In contrast, TIP3P-Ew and OPC3 display a sharper monotonic increase in free energy, suggesting stronger interfacial exclusion.

4. Conclusions

This study presented a systematic force field assessment of eight three-site rigid water models to investigate the phase equilibria, interfacial properties, and thermodynamic stability of a menthol: dodecanoic acid (3:1) hydrophobic eutectic solvent. The results demonstrate that the representation of the ES-water interface and internal water solubility is highly sensitive to the electrostatic parameterisation and Lennard-Jones settings of the selected water model. Among the models examined, OPC3 and TIP3P-ST provided the closest agreement with the experimental solubility benchmark of approximately 1.1 wt%. In contrast, TIP3P-FB significantly overestimated water uptake, particularly near the interfacial region. The nearly fourfold variation in predicted water uptake between TIP3P-FB (7.10 wt%) and TIP3P (2.00 wt %) at 303 K highlights the strong sensitivity of hydrophobic ES simulations to water model selection. From a process design standpoint, such differences could meaningfully influence predicted extract purity, solvent water loading, and downstream regeneration requirements in practical liquid-liquid extraction applications, underscoring the

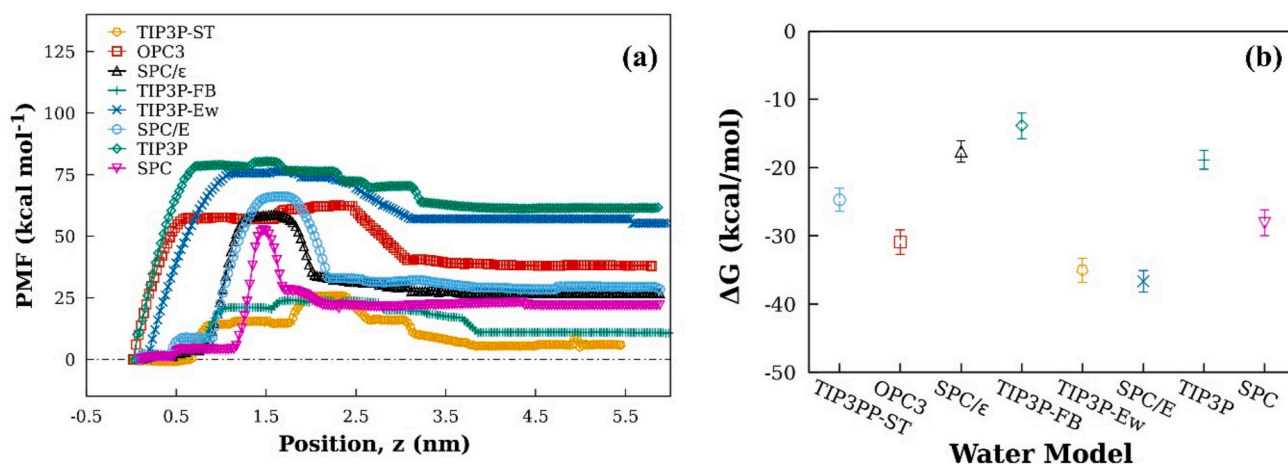


Fig. 9. (a) PMF profiles for transferring a water molecule from the ES phase to the bulk water phase along the x-direction. (b) ΔG for transfer from the ES interior to bulk water.

importance of careful water model selection when using molecular simulations as a tool for ES process development. This sensitivity arises primarily from differences in charge distribution and the resulting effective dipole moments. Models with higher cohesive energy, such as OPC3 and SPC/E, exhibit stronger water-water interactions and higher interfacial tension values exceeding 25 mN m^{-1} , which limit water dispersion into the hydrophobic ES interior and preserve a sharper phase boundary. Structural analysis based on RDF and hydrogen bond dynamics further clarified the molecular origins of these trends. Dodecanoic acid forms specific hydrogen bonding interactions with water through its acidic hydroxyl group, whereas menthol primarily contributes to the overall hydrophobic character of the solvent matrix.

This interplay produces a metastable adsorption state at the interface, consistent with the shallow local minima observed in the PMF profiles. The resulting profiles confirm that this transfer is thermodynamically favourable for all models, with $\Delta G_{\text{ES}} \rightarrow \text{water} < 0$. However, models characterised by stronger water-water cohesion, including SPC/E and TIP3P-Ew, display substantially elevated free energy plateaus within the ES interior of approximately 75 to 80 kcal/mol. This reflects a pronounced energetic penalty for water confinement in a low dielectric, hydrogen bond-disrupted environment. Conversely, the smaller magnitude of ΔG and reduced PMF barrier predicted by TIP3P-FB explain its tendency to overpredict water stabilisation within the ES phase. In conclusion, for simulations requiring accurate prediction of phase equilibria and solubility limits in hydrophobic eutectic solvent systems, OPC3 and TIP3P-ST offer the most balanced performance. The present findings demonstrate that water model selection is a decisive factor governing predicted interfacial thermodynamics, structural organisation, and transport behaviour, and must therefore be explicitly justified in computational studies aimed at designing eutectic solvent-based separation processes.

CRediT authorship contribution statement

Raj Akshat: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anand Bharti:** Writing – review & editing, Supervision, Software, Project administration, Methodology. **Padmini Padmanabhan:** Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors confirm that there are no financial or personal

relationships that could inappropriately influence or bias the work presented in this manuscript.

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Data availability

No data was used for the research described in the article.

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